

WEEK 3

FINANCIAL DATA IMPORT, TRANSFORMATION & TIME INTELLIGENCE

Introduction :

This project focuses on importing, transforming, and analyzing Shopify data using Power BI. The project uses daily stock data to analyze open price, close price, high price, low price, trading volume, and daily returns. A separate Date Table and Measures Table were created for better data organization and time-based analysis.

Objectives :

The main objectives of this project are:

- To import and prepare Shopify stock market data in Power BI.
- To create a separate Date Table for time-based analysis.
- To create calculated date columns.
- To create DAX measures for financial analysis.
- To apply time-intelligence functions.
- To create a Year–Quarter–Month–Day hierarchy.
- To use Drill Down for time-based analysis.
- To create interactive cards, charts, and slicers.
- To analyze Shopify stock price and trading volume over time.

Dataset Description :

The dataset used in this project is Shopify stock market data.

Data Table Name: SHOP_2015-05-21_2025-03-16

The main columns used in the analysis are:

- Date
- Open
- High
- Low
- Close
- Volume

The dataset contains daily stock information that was used to create financial measures and visualizations.

Data Model :

Three tables were used in the Power BI project:

1. **SHOP_2015-05-21_2025-03-16** – Main stock data table
2. **Date Table** – Used for date and time-based analysis
3. **Measures Table** – Used to organize the DAX measures

A relationship was created between the Date Table and the main stock data table using the Date column.

Power BI Data Model showing the Stock Data Table, Date Table, and Measures Table

date	open	high	low	close	adj_close	volume	Date.1
21-05-2015 00:00:00	2.79999995231628	2.87400007247925	2.41100001335144	2.56800007820129	2.56800007820129	123039000	21 May 2015
22-05-2015 00:00:00	2.60700011253357	3.109999989509583	2.59999990463257	2.83100008964539	2.83100008964539	28412000	22 May 2015
26-05-2015 00:00:00	2.98000001907349	3.03399991989136	2.9079999237061	2.96499991416931	2.96499991416931	8202000	26 May 2015
27-05-2015 00:00:00	3.06699991226196	3.08100008964539	2.70000004768372	2.75	2.75	7976000	27 May 2015
28-05-2015 00:00:00	2.75500011444092	2.7739999294281	2.64800000190735	2.744999988555908	2.744999988555908	4000000	28 May 2015
29-05-2015 00:00:00	2.72499990463257	2.73799991607666	2.68700003623962	2.72000002861023	2.72000002861023	2554000	29 May 2015
01-06-2015 00:00:00	2.72000002861023	2.74399995803833	2.66000008583069	2.73300004005432	2.73300004005432	2569000	01 June 2015
02-06-2015 00:00:00	2.74699997901917	2.74699997901917	2.66499996185303	2.6949999332428	2.6949999332428	3798000	02 June 2015
03-06-2015 00:00:00	2.67600011825562	2.75	2.65000009536743	2.74399995803833	2.74399995803833	3024000	03 June 2015
04-06-2015 00:00:00	2.74200010299683	2.74900007247925	2.55100011825562	2.67799997329712	2.67799997329712	3226000	04 June 2015
05-06-2015 00:00:00	2.665999988937378	2.73900008201599	2.609999989509583	2.73900008201599	2.73900008201599	1294000	05 June 2015
08-06-2015 00:00:00	2.69000005722046	2.74799990653992	2.68199992179871	2.69799995422363	2.69799995422363	2809000	08 June 2015
09-06-2015 00:00:00	2.66400003433228	2.84599995613098	2.66400003433228	2.83400011062622	2.83400011062622	3375000	09 June 2015
10-06-2015 00:00:00	2.84500002861023	2.88800001144409	2.79500007629395	2.80999994277954	2.80999994277954	2734000	10 June 2015
11-06-2015 00:00:00	2.84899997711182	3.07299995422363	2.84899997711182	3.07299995422363	3.07299995422363	4606000	11 June 2015
12-06-2015 00:00:00	3.13199996948242	3.369999988555908	3.125	3.36299991607666	3.36299991607666	8474000	12 June 2015
15-06-2015 00:00:00	3.3970000743866	3.81200003623962	3.29200005531311	3.54999995231628	3.54999995231628	16943000	15 June 2015
16-06-2015 00:00:00	3.60599994659424	3.83789992332458	3.58400011062622	3.81100010871887	3.81100010871887	11095000	16 June 2015
17-06-2015 00:00:00	3.96300005912781	4.21299982070923	3.80100011825562	3.86299991607666	3.86299991607666	15952000	17 June 2015
18-06-2015 00:00:00	3.84200000762939	3.88899993896484	3.4326000213623	3.5550000667572	3.5550000667572	11482000	18 June 2015
19-06-2015 00:00:00	3.63299989700317	3.67400002479553	3.25500011444092	3.55200004577637	3.55200004577637	9797000	19 June 2015
22-06-2015 00:00:00	3.67600011825562	3.84999990463257	3.60899996757507	3.69400000572205	3.69400000572205	7922000	22 June 2015
23-06-2015 00:00:00	3.77500009536743	3.80999994277954	3.51699995994568	3.54299998283386	3.54299998283386	2939000	23 June 2015

Date Table :

A separate Date Table was created using the CALENDAR() DAX function.

```
Date Table =  
CALENDAR(  
    MIN('SHOP_2015-05-21_2025-03-16'[date]),  
    MAX('SHOP_2015-05-21_2025-03-16'[date])  
)
```

The Date Table contains the following columns:

- Date
- Day
- Month
- Month Number
- Quarter
- Year

Date Table with calculated date columns

1 Date Table = CALENDAR(MIN('SHOP_2015-05-21_2025-03-16'[date]),MAX('SHOP_2015-05-21_2025-03-16'[date]))						
Date	Year	Quarter	Month	Month Number	Day	
21-05-2015 00:00:00	2015	Q2	May	5	21	
22-05-2015 00:00:00	2015	Q2	May	5	22	
23-05-2015 00:00:00	2015	Q2	May	5	23	
24-05-2015 00:00:00	2015	Q2	May	5	24	
25-05-2015 00:00:00	2015	Q2	May	5	25	
26-05-2015 00:00:00	2015	Q2	May	5	26	
27-05-2015 00:00:00	2015	Q2	May	5	27	
28-05-2015 00:00:00	2015	Q2	May	5	28	
29-05-2015 00:00:00	2015	Q2	May	5	29	
30-05-2015 00:00:00	2015	Q2	May	5	30	
31-05-2015 00:00:00	2015	Q2	May	5	31	
01-06-2015 00:00:00	2015	Q2	Jun	6	1	
02-06-2015 00:00:00	2015	Q2	Jun	6	2	
03-06-2015 00:00:00	2015	Q2	Jun	6	3	
04-06-2015 00:00:00	2015	Q2	Jun	6	4	
05-06-2015 00:00:00	2015	Q2	Jun	6	5	
06-06-2015 00:00:00	2015	Q2	Jun	6	6	
07-06-2015 00:00:00	2015	Q2	Jun	6	7	
08-06-2015 00:00:00	2015	Q2	Jun	6	8	
09-06-2015 00:00:00	2015	Q2	Jun	6	9	
10-06-2015 00:00:00	2015	Q2	Jun	6	10	
11-06-2015 00:00:00	2015	Q2	Jun	6	11	
12-06-2015 00:00:00	2015	Q2	Jun	6	12	

Calculated Columns :

Day

Day = DAY('Date Table'[Date])

Month

Month = FORMAT('Date Table'[Date],"MMM")

Month Number

Month Number = MONTH('Date Table'[Date])

Quarter

Quarter = "Q" & FORMAT('Date Table'[Date],"Q")

Year

Year = YEAR('Date Table'[Date])

The Month column was sorted using Month Number so that the months appear in chronological order.

DAX Measures :

A separate Measures Table was created to organize the DAX measures used in the project.

The following measures were created:

Open Price

Open Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[open])

Close Price

Close Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[close])

High Price

High Price = MAX('SHOP_2015-05-21_2025-03-16'[high])

Low Price

Low Price = MIN('SHOP_2015-05-21_2025-03-16'[low])

Trading Volume

Trading Volume = SUM('SHOP_2015-05-21_2025-03-16'[volume])

Previous Close

Previous Close =
CALCULATE(
 [Close Price],
 DATEADD('Date Table'[Date],-1,DAY)
)

Daily Return %

Daily Return % =
DIVIDE(
 [Close Price]-[Previous Close],
 [Previous Close],
 0
)

Time Intelligence Measures :

Time-intelligence functions were used to perform previous-year, year-over-year, and year-to-date analysis.

Close Price PY

Close Price PY =
CALCULATE(
 [Close Price],
 SAMEPERIODLASTYEAR('Date Table'[Date])
)

Close Price YoY %

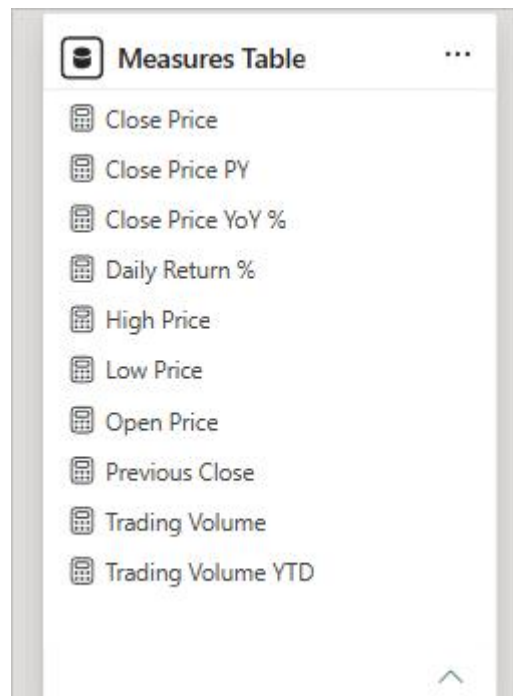
Close Price YoY % =
DIVIDE(
 [Close Price]-[Close Price PY],
 [Close Price PY],
 0
)

Trading Volume YTD

Trading Volume YTD =
TOTALYTD(
 [Trading Volume],
 'Date Table'[Date]
)

These measures were used to compare stock performance across different time periods.

DAX Measures created in the Measures Table



Date Hierarchy and Drill Down :

A date hierarchy was created using:

Year → Quarter → Month → Day

The hierarchy was used to explore the stock data at different levels of time. The Drill Down feature was used to move from Year to Quarter, Month, and Day.

Cards :

The following cards were created on the dashboard:

- Open Price
- Close Price
- High Price
- Low Price
- Trading Volume YTD

These cards provide a quick view of important financial values.

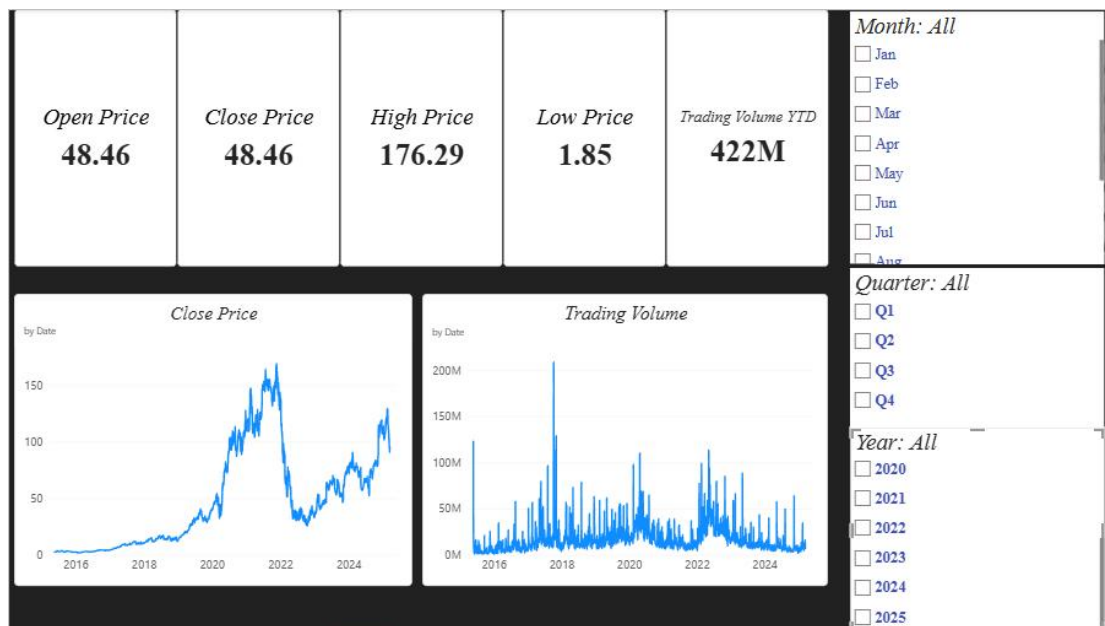
Charts and Visualizations :

Page 1

The first page contains:

- Open Price Card
- Close Price Card
- High Price Card
- Low Price Card
- Trading Volume YTD Card
- Close Price chart
- Trading Volume chart
- Year Slicer
- Quarter Slicer
- Month Slicer

Page 1 Dashboard



Page 2

The second page contains:

- Open Price chart
- Close Price chart
- Trading Volume Pie Chart

- Daily Return % chart
- Year Slicer
- Quarter Slicer
- Month Slicer

Page 2 Dashboard



Slicers and Filtering :

Three slicers were added to the dashboard:

- Year
- Quarter
- Month

The slicers allow the user to filter the stock data based on different time periods. The slicers were synchronized between the report pages.

Insights :

The dashboard provides a clear view of Shopify stock performance over time. The Open Price and Close Price visuals help to understand stock price movements.

The High Price and Low Price cards provide information about the price range. The Trading Volume visuals show changes in trading activity over time.

The Daily Return % visual helps to observe daily changes in the closing price. The time-intelligence measures provide previous-year and year-to-date analysis.

Conclusion :

This project provided practical experience in using Power BI for financial data analysis. A separate Date Table and Measures Table were created to organize the data model. DAX measures were used to calculate stock prices, trading volume, previous close, daily returns, previous-year values, and year-to-date values.

The date hierarchy, Drill Down, cards, charts, and slicers made the dashboard interactive and easier to analyze. Overall, the project helped in understanding data modelling, DAX calculations, time intelligence, and financial data visualization using Power BI.